

ISAAC AI CAPABILITIES – GAP ANALYSIS SUMMARY  
vs. TBS, PSPC, SSC, HC/PHAC EA Signals

CURRENT STATE: Isaac's 8 AI Capabilities

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|----------------------|------------------------|
| 1. Generative AI     | 5. Conversational AI   |
| 2. Knowledge AI      | 6. Perception AI       |
| 3. Predictive AI     | 7. Orchestration AI    |
| 4. Classification AI | 8. Decision Support AI |

FINDINGS: 7 CRITICAL GAPS + 3 RE-FRAMING NEEDS + 3 HC-SPECIFIC DOMAIN CAPABILITIES

TIER 1: CRITICAL GAPS (7) – Missing from Isaac, Signaled by GC Influencers

GAP 1.1: AI Model Management & Registry

Signaled by: PSPC, HC/PHAC EA, ISO 42001

Why critical: PSPC needs vendor declarations of model versions & rollback procedures

HC/PHAC EA needs model lineage for enterprise reuse

ISO 42001 §5.4 requires inventory & change control

Impact: PSPC vendor selection will fail without this

Recommendation: ADD as standalone capability

GAP 1.2: Explainability & Interpretability ⚠️ HIGH RISK FOR ARB

Signaled by: TBS ADM Directive §3.2, EU AI Act Art. 13, NIST AI RMF, ISO 42001 §5.3.5

Why critical: TBS ADM Directive MANDATES citizen right to explanation

EU AI Act requires transparency obligations (August 2026 deadline)

Conflating "explanation" with "human review" is not sufficient

Impact: ARB will flag this as non-negotiable

Recommendation: ADD as standalone capability (NIST-level design requirement)

GAP 1.3: Bias & Fairness Testing ⚠️ HIGH RISK FOR ARB

Signaled by: NIST AI RMF (MEASURE/MANAGE), EU AI Act Art. 10, ISO 42001 §5.3.3, WHO

Why critical: EU AI Act requires pre-deployment bias testing

Isaac assumes fairness without explicit testing & monitoring

WHO signals health AI cannot perpetuate inequities

Impact: ARB will flag as mandatory for high-risk AI

Recommendation: ADD as standalone capability (pre-deployment gate)

#### GAP 1.4: Enterprise Integration & API Governance

Signaled by: PSPC (vendor contracts), HC/PHAC EA (platform reuse), ISO 42001

##### §5.4.3

Why critical: PSPC vendors need clear integration specs, SLA contracts

HC/PHAC EA needs standardized HAIL interfaces

ISO 42001 requires documented interfaces & integration controls

Impact: Without this, PSPC procurement will stall; HAIL reusability fails

Recommendation: ADD as standalone capability (includes OpenAPI spec, SLAs, rate limiting, audit hooks)

#### GAP 1.5: Real-time & Streaming AI

Signaled by: HC/PHAC EA (Surveillance cluster), TBS ADM Directive (timely decisions), WHO

Why critical: HC/PHAC "Surveillance" cluster requires real-time anomaly detection (not batch)

ADM Directive implies sub-minute escalation to human review

WHO guidance: outbreak detection must happen within hours

Impact: Batch-only forecasting insufficient for health surveillance mission

Recommendation: ADD as standalone capability (event streams, sub-100ms inference, low-latency audit)

#### GAP 1.6: Data Privacy & Isolation (AI-specific)

Signaled by: SSC (cloud infrastructure), HC/PHAC EA (Protected B isolation), EU AI Act, NIST

Why critical: SSC hosts multi-tenant AI; needs federated learning & differential privacy

HC/PHAC health data cannot leak to other tenants

EU AI Act Art. 10 requires data minimization

Impact: Without this, Protected B AI workloads cannot deploy to shared cloud

Recommendation: ADD as standalone capability (federated learning, differential privacy, secure enclaves)

#### GAP 1.7: Adversarial Robustness & Security Testing

Signaled by: NIST AI RMF (robustness/security), ISO 42001 §5.3.4, SSC (cloud security)

Why critical: NIST requires testing for adversarial examples, prompt injection, data poisoning

ISO 42001 requires AI-specific security risk testing

SSC cloud infrastructure must defend against model extraction attacks

Impact: Production AI exposed to adversarial attacks without pre-deployment testing

Recommendation: ADD as standalone capability (FGSM/PGD attacks, prompt

injection, red teams)

#### ARB RISK ASSESSMENT FOR TIER 1:

-  HIGH RISK (ARB will block): Gaps 1.2, 1.3 (Explainability, Bias)
-  MEDIUM RISK (ARB will challenge): Gap 1.1 (Model Management)
-  OPERATIONAL RISK: Gaps 1.4, 1.5, 1.6, 1.7

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| TIER 2: RE-FRAMING NEEDS (3) – Isaac Capabilities Incomplete, Need Expansion |
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#### GAP 2.1: Decision Support AI → Add "Auditability & Logging" Sub-capability

Current framing: "Recommendations, prioritization"

TBS ADM Directive requires: Decision audit trail, appeal records, regulatory disclosure

Revision needed: Add explicit "Auditability & Logging" covering TBS §3.1 mandates

#### GAP 2.2: Orchestration AI → Add "Change Management & Rollback" Sub-capability

Current framing: "Agentic workflows, task automation"

PSPC/HC/PHAC need: Staged rollout, automated rollback triggers, regression testing

ISO 42001 requires: Change control procedures (§5.4.7)

Revision needed: Expand to include safe deployment patterns

#### GAP 2.3: Perception AI → Split into "Structured" and "Unstructured" Variants

Current framing: "OCR, vision, speech processing" (lumped together)

HC/PHAC EA signals: Document digitization (99%+ accuracy) ≠ lab image analysis (bias testing required)

Governance profiles: Different risk/control needs per variant

Revision needed: Split into two capabilities with different governance profiles

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| TIER 3: HC/PHAC DOMAIN SPECIALIZATIONS (3) – Not GC-wide, but HC-Strategic |
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#### GAP 3.1: Scientific Computing AI (HC-specific)

Signaled by: HC/PHAC EA, PMRA, WHO AI Guidance

Scope: Lab collaboration, literature synthesis, hypothesis generation, experiment design

Status: Likely composite of Knowledge AI (RAG) + Generative AI (synthesis) applied to science domain

Recommendation: Consider separate "HC/PHAC Domain Specializations" section (not core GC framework)

#### GAP 3.2: Epidemiological Signal Detection AI (HC-specific)

Signaled by: HC/PHAC EA "Scale and Strengthen Surveillance", Surveillance cluster, WHO

Scope: Real-time anomaly detection, outbreak prediction, signal quality assessment

Status: Composite of Predictive AI + Real-time AI + Data Privacy; mission-critical for HC

Recommendation: Consider separate "HC/PHAC Domain Specializations" section

### GAP 3.3: Regulatory Intelligence AI (HC-specific)

Signaled by: PMRA, HC AI Strategy "Enhance Regulatory Processes", HC/PHAC procurement

Scope: Literature triage, PDF data extraction, regulatory requirement matching

Status: Composite of Classification AI + Perception AI + Knowledge AI; PMRA procurement priority

Recommendation: Consider separate "HC/PHAC Domain Specializations" section

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## REVISED ISAAC FRAMEWORK (8 → 15 Core + 3 HC-Specific)

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### EXISTING (keep as-is):

- └ Generative AI ✓
- └ Knowledge AI ✓
- └ Predictive AI ✓
- └ Classification AI ✓
- └ Conversational AI ✓
- └ Perception AI ✓ (split)
- └ Orchestration AI ✓ (expand)
- └ Decision Support AI ✓ (expand)

### NEW (add immediately):

- └ 1.1: AI Model Management & Registry
- └ 1.2: Explainability & Interpretability ⚠ ARB priority
- └ 1.3: Bias & Fairness Testing ⚠ ARB priority
- └ 1.4: Enterprise AI Integration
- └ 1.5: Real-time & Streaming AI
- └ 1.6: Data Privacy & Isolation
- └ 1.7: Adversarial Robustness Testing

### PLUS EXPANSIONS (Gap 2):

- └ Decision Support AI + Auditability & Logging sub-capability
- └ Orchestration AI + Change Management & Rollback sub-capability
- └ Perception AI split into Structured (OCR) & Unstructured (vision)

### OPTIONAL HC SPECIALIZATIONS (Gap 3):

- └ Scientific Computing AI (domain-specific: lab + research)
- └ Epidemiological Signal Detection AI (mission-critical: surveillance)
- └ Regulatory Intelligence AI (PMRA procurement: literature + regulatory)

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## IMMEDIATE ACTIONS FOR ARB SUBMISSION

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#### BLOCKING ISSUES (fix immediately):

- Gap 1.2 (Explainability): Define governance profile anchored to TBS ADM Directive §3.2
- Gap 1.3 (Bias Testing): Add pre-deployment testing gate + continuous monitoring profile
- Gap 1.1 (Model Management): Define model registry, versioning, rollback procedures

#### CRITICAL FOR ARB CREDIBILITY:

- Gap 1.4 (Integration): Document HAIL interface specs, SLA guarantees, audit hooks
- Gap 2.1 (Auditability): Expand Decision Support AI with TBS ADM §3.1-compliant logging
- ISO 42001 alignment: Trace all gaps back to ISO 42001 control references

#### OPERATIONALLY IMPORTANT (add before v2.4 finalization):

- Gaps 1.5, 1.6, 1.7: Define governance profiles for real-time, privacy, robustness

#### NICE-TO-HAVE (after core ARB approval):

- Gaps 3.1-3.3: Build HC/PHAC domain specializations as separate section

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#### SPREADSHEET UPDATE PLAN

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##### Current sheets:

- ✓ Isaac AI Capabilities (8 rows) → UPDATE to 15 rows + Gaps 2.1-2.3 expansions

##### Add new sheets:

- Gap 1.1-1.7: AI Model Management & Registry (with governance profile, policy anchor, HAIL integration)
- Gap 1.2: Explainability & Interpretability (with governance profile, TBS ADM reference, examples)
- Gap 1.3: Bias & Fairness Testing (with governance profile, EU AI Act reference, metrics)
- Gap 1.4: Enterprise AI Integration (with governance profile, SLA templates, API specs)
- Gap 1.5: Real-time & Streaming AI (with governance profile, HC Surveillance use cases)
- Gap 1.6: Data Privacy & Isolation (with governance profile, Protected B requirements)
- Gap 1.7: Adversarial Robustness Testing (with governance profile, SSC security requirements)
- HC/PHAC Domain Specializations (3 rows: Scientific Computing, Epidemiological Signal Detection, Regulatory Intelligence)

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## POLICY ANCHORS BY STAKEHOLDER

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### TBS SIGNALS (Mandatory):

Gap 1.2 (Explainability) → GC ADM Directive §3.2 (citizen right to explanation)

Gap 2.1 (Auditability) → GC ADM Directive §3.1 (decision records, appeal method)

### PSPC SIGNALS (Procurement):

Gap 1.1 (Model Management) → Vendor capability declarations, rollback procedures

Gap 1.4 (Integration) → API specs, SLA contracts, rate limiting, monitoring hooks

### SSC SIGNALS (Infrastructure):

Gap 1.6 (Data Privacy) → Protected B isolation, multi-tenant security

Gap 1.7 (Robustness) → Cloud security, adversarial attack defense

### HC/PHAC EA SIGNALS (Enterprise Readiness):

Gap 1.5 (Real-time AI) → Surveillance cluster real-time requirements

Gap 1.1 (Model Management) → Platform reuse & enterprise AI readiness

Gap 1.4 (Integration) → HAIL standardized interfaces

### INTERNATIONAL STANDARDS (Compliance):

ISO 42001: Gaps 1.1, 1.2, 1.3, 1.4, 1.6, 1.7

EU AI Act: Gaps 1.2, 1.3, 1.6 (August 2026 deadline)

NIST AI RMF: Gaps 1.2, 1.3, 1.5, 1.6, 1.7

WHO AI Guidance: Gaps 1.3, 1.5 (health equity)

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FULL DETAILED ANALYSIS: See gap\_analysis.md (delivered separately)